

Unveiling Cognitive Attacks on Multimodal LLM-based XR Cognitive Assistants through Trojans

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Abstract

Extended Reality (XR), empowered by large vision-language models (LVLMs), rapidly transforms user interactions through complex tasks by overlaying visual cues and textual instructions, enhancing performance in daily and high-stakes scenarios. However, this growing reliance also introduces new, underexplored security risks, particularly those that subtly target users' perception and cognition rather than system infrastructure. In this paper, we introduce, for the first time, a cognitive attack Trojan that targets LVLM-guided intelligent XR cognitive assistants by subtly manipulating systems' perception to degrade user task performance. Unlike other attacks, this Trojan remains hidden within the application and does not interfere with the application's functionality until the attack is triggered. Once triggered, such attacks increase users' cognitive load, extend task completion times, and disrupt the immersive flow of the XR experience, leading to frustration and reduced trust in the intelligent XR cognitive assistant. To demonstrate the effectiveness of this attack, we first develop *CogXR*, an LVLM-guided intelligent XR cognitive assistant on the Magic Leap 2 headset that provides real-time visual and textual guidance for diverse cognitive tasks. We then evaluate *CogXR* across two distinct cognitive task categories: spatial reasoning tasks (e.g., Rubik's Cube solving) and procedural tasks (e.g., cooking) to assess how the attack influences performance under varying levels of cognitive demand. Our findings demonstrate that, when triggered, the Trojan subtly manipulates visual input, misleading the LVLM across both task categories, resulting in prolonged task completion times, increased cognitive demand, and disrupted immersive flow. **Our conducted user studies on both spatial reasoning and procedural tasks show that the cognitive attack Trojan increases task completion time by up to 43% on the spatial-reasoning task and 25.2% on the procedural task relative to the no-attack condition**, elevating cognitive load and degrading performance, highlighting critical vulnerabilities in XR systems.

CCS Concepts

- **Security and privacy** → **Software and application security**;
- **Human-centered computing** → *Mixed / augmented reality*; • **Computing methodologies** → Artificial intelligence.

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Keywords

Extended Reality, Cognitive Attack, Large Vision Language Models, Cognitive Load.

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1 Introduction

The recent trend of integrating advanced pre-trained large vision-language models (LVLMs) into extended reality (XR) systems is revolutionizing their capabilities, enabling personalized, context-aware interactions that significantly reshape our engagement with the world. **One such notable application is LVLM-driven intelligent XR systems, denoted as intelligent XR cognitive assistance, which overlay step-by-step instructions onto users' fields of view to guide them through complex tasks. For instance, spatial reasoning tasks (e.g., puzzle and configuration solving) and procedural tasks (e.g., assembly, maintenance, and cooking) are significantly enhanced by LVLMs in XR environments [21, 22, 34, 49, 50, 54, 60, 66, 68, 76].**

Furthermore, researchers have also integrated LVLMs into XR systems to facilitate nuanced tasks such as pronoun disambiguation for voice assistants [35], intelligent XR-based activity recording and analysis [27], documentation and writing support [11], augmented object intelligence [20], understanding user behaviors and contextual reasoning [29], and even harmful content detection within XR environments. These advancements underscore the role of LVLMs in transforming XR into a robust cognitive augmentation platform that supports users in both mundane and mission-critical scenarios. Beyond this, such integrations are also expected in *safety-critical applications*, such as the recently announced next-generation military XR headsets, designed and built by Anduril and Meta to provide warfighters with enhanced perception, assistance, and enable intuitive control of autonomous platforms on the battlefield [5].

As LVLMs increasingly permeate daily life, they also pose significant threats to users' safety, security, and privacy [3, 16, 31]. *However, since leveraging LVLM to XR is still a new concept, the vulnerabilities of such integrations are yet vastly underexplored.* For instance, a slight alteration in an object's appearance or insertion of an adversarial element can cause the LVLM to misidentify critical features or misunderstand user intent [69, 71, 72, 74]. Unfortunately, as research on the typical security and privacy consequences of XR and LVLM integration grows, the *cognitive security* [18, 19] of this integration for intelligent XR cognitive assistants has received little attention from the community. Indeed, cognitive security is critical across many applications, as the state of the user's cognition is directly related to stress, task performance, decision-making, and other factors crucial to applications such as public safety and

national defense. For instance, in a battlefield situation, the compromise of a soldier's cognitive security can lead to catastrophic consequences [18]. Commercial XR systems apply cognitive engineering principles during system development; however, due to the product supply chain, it is not always possible to ensure that intelligent XR cognitive assistants operate safely when facing an adversary intent on interfering with a mission. This is especially true if the threat is dormant, e.g., a *Trojan*, which can only be activated externally by a trigger (e.g., a visual trigger). Since Trojans are by definition dormant, stealthy, and stay inactive during normal operation, it is extremely challenging to detect Trojans in AI (TrojanAI) [53] during the software or hardware testing phases. Hence, there is a pent-up need to explore the impact of cognitive security vulnerabilities of intelligent XR cognitive assistants. *It is worth noting that the goal of a cognitive attack differs from traditional Trojans, which compromise users' security (e.g., by stealing passwords, credit card numbers, etc.) and may go far beyond. For instance, jeopardizing the decision-making of intelligent LVLM XR assistants by triggering a Trojan during firefighting, search-and-rescue, or battlefield operations can significantly impair the cognitive capabilities of first responders and soldiers, putting human lives at stake.*

This paper introduces a novel cognitive attack Trojan (CAT) targeting LVLM-guided intelligent XR cognitive assistants. Inspired by TrojanAI [53] and hardware Trojan attacks [8, 23, 30], which have long been a well-known threat in the AI and hardware security domains, these Trojans remain dormant until a specific external trigger condition is met. Once activated, they execute hidden malicious behaviors with subtle, hard-to-detect consequences, often severely affecting the system's functionality and security. Similarly, our proposed CAT remains dormant and undetectable during normal XR system operation, causing no disruption to the user experience. However, when a predefined condition is met, the Trojan becomes active (triggered) and manipulates the intelligent XR system's behavior in a manner imperceptible to the user. This results in significantly increased task completion times, elevated user cognitive load, and disruption of the immersive flow of the XR experience. To elucidate our proposed idea, we first develop a real-time **baseline** LVLM-guided intelligent cognitive assistant framework **CogXR** and implement it on the Magic Leap 2 XR headset to provide step-by-step guidance for representative spatial reasoning tasks (i.e., Rubik's Cube solving) and procedural tasks (i.e., cooking), as shown in Figure 1. CogXR processes captured frames from the XR camera using an LVLM to provide step-by-step guidance through visual overlays and concise textual instructions. We show that CogXR works as expected in an ideal environment (no attack launched), which is referred to as the benign environment and serves as the baseline in the paper. Next, to demonstrate the effectiveness of the proposed CAT, we develop a malicious version of the CogXR for the same tasks and perform a user study using the benign and malicious CogXR to analyze key performance metrics, including task completion time and cognitive load, as measured by the NASA Task Load Index (TLX) [25]. *Our findings reveal that the proposed cognitive attack substantially increases task completion time across both spatial-reasoning (Rubik's Cube solving) and procedural tasks (cooking), with average increases of 43% and 25.2%, respectively, compared with benign conditions, thereby elevating cognitive load and degrading task performance. To the best of our knowledge, this*

is the first work to propose cognitive attacks against LVLM-guided intelligent XR cognitive assistants. We believe our work will help XR developers better understand the vulnerabilities of such systems and pave the way for appropriate measures to mitigate them.

2 Related Works

LVLM for cognitive tasks in XR: Recent advancements in pre-trained LVLMs (e.g., GPT-4o [2], Google Gemini [63]) have advanced multimodal understanding, and their zero-shot inference integrates naturally into XR to enrich immersive experiences. Prior research demonstrates their effectiveness in real-time, context-aware guidance across both spatial reasoning (e.g., Rubik's Cube and Sudoku solving) [21, 24, 50, 54] and procedural tasks (e.g., cooking, furniture assembly, Coffee making) [10, 22, 34, 40, 60, 66]. A notable application is LVLM-guided intelligent XR cognitive assistance, which overlays step-by-step instructions onto users' fields of view [49, 60, 76]. For instance, Srinidhi et al. [60] showed that integrating LVLMs with physical-world perception enables context-aware guidance (e.g., coffee-making, furniture assembly), and such systems are now deployed at scale, exemplified by Meta's Ray-Ban Smart Glasses [67]. However, as LVLMs become embedded in high-stakes XR assistants, they introduce critical security risks. For instance, Zhang et al. [74] demonstrated attacks through environment manipulation, visual overlays, and prompt modification; however, such attacks rely on easily detectable physical perturbations and face barriers in restricted API environments [7, 26, 38], limiting their effectiveness against real-world systems. In contrast, semantic-level attacks that degrade LVLM visual-semantic reasoning without visible traces remain unexplored, yet could imperceptibly alter task guidance and cause users to make critical errors in consequential tasks (e.g., cooking). This motivates a new kind of stealthy, impactful attack that manipulates LVLM perception to degrade user cognitive load and task performance.

Safety and Security Risks in XR: XR systems are rapidly advancing within human-computer interaction HCI, blurring the boundary between real and virtual environments [17, 60], and with lighter, more powerful displays (e.g., Microsoft HoloLens [74], Magic Leap [60]), research is moving toward intelligent, context-aware XR. This integration, however, has expanded XR's attack surface, exposing diverse threats including PMA, memory tampering, keystroke inference, side-channel exploits, human-joystick control, and overlay attacks [3, 4, 9, 12–14, 31, 41, 43, 46, 55, 57, 58, 61, 64, 65, 72] that disrupt immersion and degrade users' cognitive state [19]. For instance, Tseng et al. [65] showed that multisensory manipulation can mislead decisions and cause physical harm, while Cheng et al. [14] reported delayed reactions under vision- and audio-based attacks. Similarly, sensor spoofing [13], GPU disruptions [43], and UI vulnerabilities [58] compromise XR reliability, and combining obstruction with misinformation prevents critical object perception [72]. Meanwhile, Xiu et al. [69] introduced AR-VIM to detect non-occlusive visual manipulation. To counter such threats, researchers have explored LVLMs for detecting critical visual cues [26, 71]. However, these models remain susceptible to adversarial manipulation, where even subtle visual alterations cause misidentification, incorrect instructions, and reduced system reliability.

3 Proposed Cognitive Attack Trojan

In the following sections, we present the core idea of the proposed attack, along with its threat model, attack scenario, and methodology for attack modeling and execution. Details are provided below.

3.1 Hardware Trojan vs. Cognitive Attack Trojan

Hardware Trojan attacks are a well-established threat in the hardware security domain [8, 23, 30], where a Trojan is covertly embedded into a circuit or chip during the design or fabrication process. It remains dormant until a specific external condition, known as the **trigger** (e.g., a particular input signal or circuit state), is detected, upon which it executes a hidden malicious action, known as the **payload** (e.g., corrupting data or disabling functionality), often with severe and hard-to-detect consequences. Inspired by this trigger–payload concept, we introduce a Cognitive Attack Trojan (CAT) for LVLM-guided intelligent XR cognitive assistant systems. Rather than targeting hardware circuits, our proposed CAT targets the intelligent XR assistant’s visual processing pipeline, where the **trigger** is an *attacker-specified* unique object (e.g., a cup with a QR code, a star-shaped sticker, or specialized text). Importantly, the trigger is not activated by arbitrary objects in the environment; only the exact object with a **unique, predefined characteristic** chosen by the attacker causes activation, making the Trojan highly selective and stealthy. The **payload** consists of a small code segment that intercepts the current camera frame upon trigger activation and applies a targeted semantic-level manipulation (e.g., recoloring a Rubik’s Cube facelet, shifting an ingredient’s visual properties, etc.) before forwarding it to the LVLM.

3.2 Cognitive Attack Trojan Concept

LVLM-guided intelligent XR cognitive assistant systems rely heavily on visual perception, real-time frame capture, and accurate environment understanding. To build such systems, developers commonly integrate SOTA open-source third-party dependencies (e.g., libraries, packages, and assets), such as OpenCV for Unity [59] for real-time image processing and OpenAI for Unity [51] for LVLM API access. An attacker can exploit this trust by publishing malicious third-party dependencies on popular platforms such as GitHub or the Unity Asset Store. Once integrated, any of these dependencies can covertly embed the CAT (see Section 3.1), which activates silently at runtime and semantically manipulates the camera frame (e.g., flipping digits, or changing colors) before LVLM inference. Since the component genuinely delivers its advertised functionality, developers integrate it without suspicion. Moreover, since the CAT remains dormant in the absence of the trigger object, it is extremely difficult for testers or users to detect. Furthermore, exploiting the egocentric nature of XR environments, the visual trigger can be introduced by any individual aware of it, not necessarily the original attacker, allowing multiple actors to independently activate the dormant CAT and creating a uniquely stealthy, difficult-to-detect attack vector.

3.3 Threat Model

Attacker Goals: The attacker’s primary objective is to subtly degrade user task performance within LVLM-guided intelligent XR

cognitive assistant systems. These systems provide XR-based multimodal guidance across a range of real-world procedural and spatial-reasoning tasks [21, 22, 34, 50, 54, 56, 60, 66, 76]. By manipulating selected visual inputs, the adversary causes the LVLM to generate erroneous yet contextually plausible instructions, making tasks harder to complete while the system appears to function normally. The impact varies significantly across task contexts. In everyday tasks (e.g., cooking, Rubik’s Cube solving, furniture assembly), the attack induces incorrect actions that disrupt task flow, increase task completion time and cognitive demand, and elevate user frustration. In mission-critical operations, such as hazard localization in firefighting, route planning in search-and-rescue, and tactical procedure execution on battlefields [56], the consequences become far more severe, as impairing the decision-making of intelligent XR assistants can critically compromise the situational awareness of firefighters, paramedics, and soldiers, directly putting human lives at stake. Across both contexts, the attack increases confusion, prolongs task completion, and elevates cognitive effort, all while remaining undetected.

Attacker Profile and Capabilities: We assume the adversary can publish a malicious library (with an embedded CAT) on widely used distribution platforms (e.g., GitHub, Unity Asset Store). The library provides genuine, useful functionality for XR development (e.g., camera frame preprocessing, LVLM API call batching), making it attractive to developers building LVLM-guided intelligent XR cognitive assistants. This attack model follows prior work that weaponized trusted software dependencies to inject malicious code into applications that rely on them [33, 44], and requires only general knowledge of how egocentric frames are captured, preprocessed, and forwarded to the LVLM for reasoning [2, 49, 60, 76]. As discussed in the Attack Concept (Section 3.2), the library’s legitimate frame-processing role grants the CAT natural access to the camera-to-LVLM pipeline while raising no suspicion, especially since modern XR applications rely on numerous third-party dependencies (e.g., OpenCV for Unity [59], platform SDKs) whose source code is rarely fully audited [33]. Leveraging this access, the adversary can design minimal yet effective manipulations (e.g., subtle color shifts, label swaps, etc.) that mislead the LVLM while remaining imperceptible to both the developer and the user. Notably, the attack can be carried out by a single actor who develops the malicious library and later activates it, or by two independent individuals, with one embedding the CAT and the other triggering it by placing the predefined object in the camera’s view, without any coordination between them. *The beauty of this attack is that once the application is deployed with the CAT embedded, it can be launched at any time later by the adversary (or any individual aware of the attacker-specified trigger object) without accessing the headset, hardware, software, or network stack, only by placing the predefined object in the XR’s camera frame.*

3.4 CAT Modeling and Execution

We model a CAT that leverages runtime manipulation of egocentric frames to generate incorrect instructions in LVLM-guided intelligent XR cognitive assistant systems. The XR system captures egocentric frames from the environment via a head-mounted camera and forwards them to the LVLM, which interprets the visual

context and generates step-by-step multimodal guidance for task completion. These systems support a wide range of cognitively guided tasks, such as spatial reasoning tasks (e.g., Rubik’s Cube solving, Sudoku solving, etc.), and procedural tasks (e.g., cooking, making coffee, assembling furniture, etc.) [21, 22, 34, 49, 50, 54, 60, 66, 68, 76]. To model frame manipulation, we represent the semantically relevant portion of the scene as a matrix $F_k \in C^{n \times m}$, where $F_k(i, j)$ denotes the semantic label or class assigned to a localized region at row i and column j . C is the set of all valid label values (e.g., object classes, colors, ingredient types, etc.) n and m refer to the number of rows and columns (spatial coordinates), and k denotes the frame index in the sequence. As discussed in the threat model, the attacker has flexibility in which element to perturb. For instance, in spatial reasoning tasks (e.g., Rubik’s Cube solving, Sudoku solving), each element in F_k represents a semantic attribute of a scene object (e.g., color, shape, position). In procedural tasks (e.g., cooking), elements in F_k may denote object properties such as categories, states, or quantities (e.g., ingredients, utensils, etc.). A single position $(i^*, j^*) \in \{0, \dots, n-1\} \times \{0, \dots, m-1\}$ is dynamically selected from the segmentation mask of the payload-specified target object, where i^* and j^* are spatial coordinates identifying the targeted semantic element in the scene. The attack matrix F_k^{att} is defined as:

$$F_k^{att}(i, j) = \begin{cases} \delta(F_k(i^*, j^*)) & \text{if } (i, j) = (i^*, j^*) \\ F_k(i, j) & \text{otherwise} \end{cases}$$

Where δ replaces the original label at $F_k(i^*, j^*)$ with a different valid label from $C \setminus \{F_k(i^*, j^*)\}$, ensuring minimal perturbation while maximizing LVLM misinterpretation.

At runtime, the system continuously detects visual entities from each frame, storing them in a set $R = \{r_1, r_2, \dots, r_n\}$, and evaluates each $r_i \in R$ against a Boolean Trojan trigger function $\phi(r_i) \in \{0, 1\}$. If no entity satisfies $\phi(r_i) = 1$, clean frames are forwarded to the LVLM unchanged. Once any $r_i \in R$ satisfies $\phi(r_i) = 1$, the Trojan activates at each subsequent decision step: it selects the most recent frame, extracts F_k , dynamically localizes the payload-specified target object, applies δ at the corresponding coordinate (i^*, j^*) , and re-embeds the modified matrix before forwarding the manipulated frame sequence F' to the LVLM $\mathcal{M}$. This perturbation persists across all subsequent decision steps as long as the trigger condition remains satisfied. Although only a single element is altered, this subtle modification distorts the model’s scene interpretation, resulting in incorrect guidance that degrades task performance, increases cognitive load, and prolongs task-completion time in LVLM-guided intelligent XR cognitive assistant systems. *For the complete execution algorithm and a representative implementation of the embedded Trojan, please refer to the supplementary material.*

3.5 Novelty of the Proposed Attack

Our proposed CAT operates fundamentally differently from SOTA multimodal prompt injection, adversarial, and classical backdoor attacks on LVLMs, as well as XR-specific attacks that do not target LVLM-guided cognitive assistants [15, 37, 45]. Prompt injection embeds malicious instructions in text or visual patterns (e.g., concealed text or disruptive overlays) [15, 37]; adversarial attacks craft

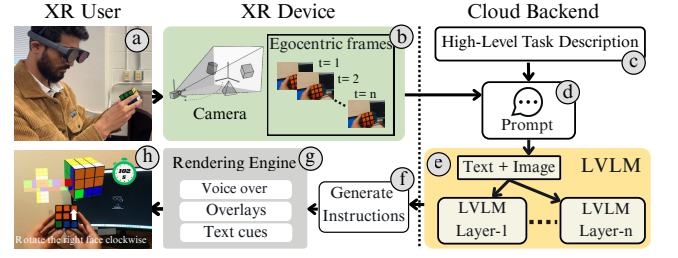


Figure 1: An overview of the *baseline CogXR* system. (a–b) An XR user performs a spatial reasoning task (Rubik’s Cube solving) while the device captures egocentric frames in real-time. (c–e) The system constructs a prompt from the captured frames and task description, and sends it to the LVLM for context-aware guidance generation. (f–h) The rendering engine delivers the generated instructions to the user via visual overlays and textual cues.

pixel-level perturbations from knowledge of the model’s parameters or architecture [7, 37, 73, 75]; and classical backdoors poison training data to embed static patterns into model weights [36, 39]. All three require either training-time access or model internals, and are constrained by commercial LVLMs being reachable only through restricted cloud APIs [7, 45]. In contrast, our CAT requires no API access, model internals, training-data poisoning, or weight modification; instead, the Trojan activates at runtime by altering object-level properties via the attacker-specified visual trigger, remaining effective regardless of the underlying LVLM architecture. Beyond those attacks, XR-specific attacks operate at observable or scene-level boundaries rather than at the LVLM’s semantic reasoning. For instance, Dirksen et al. [19] address cognitive security in AR through attention-based manipulation yet operate at the scene-level attention layer rather than the LVLM semantic reasoning level. Meanwhile, obstruction attacks overlay virtual content to occlude the real world, creating readily detectable artifacts [70]. On the other hand, AR-VIM [69] alters AR overlays but only changes what users see, not the internal object representations LVLMs process. Similarly, Zhang et al. [74] manipulate query timing, environment, overlays, and prompts at observable input/output boundaries, making them detectable via multimodal verification [26] and LVLM safety filters [38]. In contrast, our proposed CAT modifies minimal visual elements (e.g., color, size, or labels) before LVLM inference, targeting semantic-level object representations rather than the observable channels existing defenses monitor. *Moreover, the Trojan stays dormant until triggered by anyone aware of the attacker-specified object (not necessarily its implanter), making it extremely difficult for software testers to detect, and once implanted, it can be enabled or disabled without accessing the system.*

4 Baseline: CogXR System Design

CogXR is an intelligent system that assists users in performing complex, multi-step physical tasks by integrating the spatial perception of XR headsets with the semantic and visual reasoning of LVLMs, as shown in Figure 1. The system continuously captures frames using front-facing RGB cameras on head-mounted XR devices (e.g., the Magic Leap 2 [74]) and selectively transmits buffered frames to an LVLM backend via remote API calls or local/edge

computing resources. For this, we use GPT-4o [2], which interprets captured frames, infers user progress, and generates context-aware, step-by-step instructions, leveraging its training on massive multimodal data to produce personalized guidance across both spatial-reasoning and procedural task categories [54, 60, 66]. Building on these capabilities and inspired by SOTA LVLM-guided intelligent XR cognitive assistants [49, 60, 76], we design CogXR as an intelligent, context-aware XR cognitive assistant using the structured Prompt 1, where *task* denotes the XR activity (e.g., Rubik’s Cube solving, cooking), *key objects* refer to task-relevant items in the egocentric view (e.g., colors, utensils), and *task-specific constraints* capture task-dependent requirements (e.g., target configuration, specifications). Using these structured inputs alongside egocentric frames, the LVLM analyzes user progress and generates precise next-action instructions delivered as text prompts, spatially anchored visual overlays, and voice instructions.

Although CogXR is applicable across a wide range of tasks, as discussed in Section 3.3, we evaluate the proposed CAT on two representative categories that capture fundamentally different cognitive demands. For *spatial reasoning*, we use Sudoku and Rubik’s Cube solving, as both require mental rotation, geometric reasoning, and pattern recognition [21, 24, 50, 54], making them well-suited for measuring performance degradation at the semantic level perturbations. For *procedural* tasks, we use steak cooking and coffee making, as both involve discrete, ordered steps and continuous adaptation to the environment state [10, 22, 40, 49, 60], representing practical, safety-relevant scenarios. Together, these four tasks provide complementary coverage of the cognitive demands typical of real-world XR cognitive assistants. Across all tasks, CogXR captures egocentric frames to extract the current task state into a semantic matrix representation F_k , instructs the LVLM to determine the next optimal action, and re-estimates F_k after each user action to verify execution and maintain task-state awareness, enabling progressive guidance toward completion. When the CAT is active, a single semantic perturbation to F_k (e.g., modifying a Sudoku digit, changing a Rubik’s facelet color, altering a steak’s color from dark to light to appear undercooked, or changing the coffee machine water level from sufficient to insufficient) corrupts the state perception, causing the LVLM to misinterpret the task configuration and generate incorrect or potentially unsafe action recommendations, as shown in Figure 2. For *task-specific visual references*, please refer to the supplementary material.

It is important to note that, in the remainder of this paper, we refer to the CogXR system operating in a benign environment as the baseline CogXR, and the CogXR system implanted with a CAT as the malicious version of CogXR, denoted malicious CogXR.

5 System Implementation and Evaluation

We implemented both baseline CogXR and malicious CogXR in Unity 2022.3.20f1, targeting the Magic Leap 2 headset running OS version 1.12.0, using the Magic Leap Unity SDK 2.6.0. The system supports multiple XR-guided task scenarios, including spatial-reasoning tasks (Rubik’s Cube and Sudoku solving) and procedural tasks (cooking and coffee making). For this, we use the pretrained GPT-4o model [2], accessed via the OpenAI API [1], as the LVLM

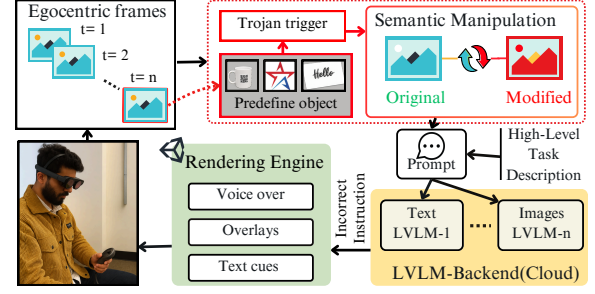


Figure 2: Overview of the CAT pipeline. XR headset frames are analyzed to detect a predefined trigger object. Upon detection, CAT alters the visual content, which makes the LVLM generate misleading instructions, which are then delivered through the XR interface.

backend to interpret egocentric visual input, estimate user progress, and generate context-aware, step-by-step multimodal guidance. For auditory feedback, we integrated Magic Leap’s text-to-speech utility to provide voice guidance during task execution. The baseline CogXR captures egocentric frames from the headset camera and uses the LVLM to deliver step-by-step guidance through visual overlays, concise text, and voice instructions. In the malicious CogXR, we simulate the attack by embedding CAT within a third-party library integrated into the application’s visual-processing pipeline, which silently alters visual input before it is forwarded to the LVLM. That comprises two synchronized phases. First, CAT continuously analyzes incoming egocentric frames to detect a predefined visual trigger. Once the trigger is detected, CAT activates the corresponding payload. The payload uses a locally deployed YOLO11-based instance-segmentation model [28] to dynamically localize the payload-specified target object, selects the highest-confidence instance above a predefined threshold, refines its segmentation mask using OpenCV for Unity [59], and applies the corresponding semantic modification, before forwarding the manipulated frame to the LVLM.

Prompt 1: Example prompt for CogXR system

You are an intelligent cognitive assistant designed to support physical task execution in XR environments. From the given *<egocentric frames>* captured by the headset’s front-facing camera, identify the *<key objects>* in view. Combine this information with the *<known information>* from the XR environment, the *<previous instruction>* issued to the user, the specified *<task>*, and its *<task-specific constraints>*. Analyze the user’s current progress and determine the next optimal action. Provide precise, context-aware natural-language instructions grounded in the user’s visual perspective to support accurate and effective task completion.

System Evaluation Results: We measure end-to-end latency from headset frame capture to rendering the generated instruction in the XR interface. As reported in Table 1, baseline CogXR requires 4.236–4.690 seconds per guidance cycle, while CAT-enabled CogXR requires 4.420–4.912 seconds. We demonstrate the flexibility of CAT trigger design by using three different predefined visual trigger objects: a white cup bearing a QR code, a star-shaped sticker, and a card displaying the text “Hello,” with only one trigger active per execution. Each trigger is evaluated over four runs (12 runs in total), evenly divided between trigger-present and trigger-absent

Table 1: Component-wise latency (in seconds) of baseline (Base) and malicious (Mal) CogXR across Rubik’s Cube solving (RC), Sudoku solving (SS), cooking (C), and coffee making (CM). Sem. denotes semantic perturbation, and N/A denotes not applicable. The aggregate Trigger Detection and CAT Processing values are means across the three independently tested trigger variants.

System Component	Spatial-Reasoning Tasks				Procedural Tasks			
	RC		SS		C		CM	
	Base	Mal	Base	Mal	Base	Mal	Base	Mal
End-to-End System	4.236	4.420	4.354	4.548	4.690	4.912	4.511	4.747
GPT-4o Backend	4.169	4.175	4.285	4.292	4.612	4.625	4.438	4.449
XR Processing	0.067	0.068	0.069	0.070	0.078	0.079	0.073	0.074
Frame Capture	0.005	0.005	0.006	0.006	0.006	0.006	0.005	0.005
Response Parsing	0.041	0.042	0.041	0.042	0.047	0.048	0.045	0.046
XR Rendering	0.021	0.021	0.022	0.022	0.025	0.025	0.023	0.023
CAT Processing		0.177		0.186		0.208		0.224
Trigger Detection		0.012		0.013		0.015		0.014
QR-Coded Cup	N/A	0.012	N/A	0.013	N/A	0.015	N/A	0.014
Star Sticker		0.010		0.011		0.013		0.012
Text Card		0.014		0.015		0.017		0.016
Sem. Perturbation		0.165		0.173		0.193		0.210

conditions. Across all runs, CAT achieved 100% trigger-detection accuracy, activating only when a predefined trigger was present, with no false positives or false negatives. Overall, CAT introduces 0.184–0.236 seconds of additional end-to-end latency, corresponding to approximately 4.34–5.23% overhead relative to baseline CogXR.

6 Cognitive Attack Trojan Evaluation

This section presents the user study design and results, demonstrating the effectiveness of our proposed CAT on the CogXR system across both task categories.

Hypotheses: The primary focus of this research is to identify and understand the impact of the CAT on the CogXR system for both procedural and spatial-reasoning tasks. Specifically, we investigate whether the attack can bias the assistant’s guidance generation, producing incorrect or inefficient instructions that significantly increase task completion time, elevate cognitive demand, and disrupt the user’s immersive experience. To systematically measure and evaluate the effects of the proposed CAT on the performance and trustworthiness of the CogXR system, we formulate the following three hypotheses as the basis for our experimental study.

- **H1:** Cognitive attack Trojan will significantly increase task completion time compared to without cognitive attack Trojan condition.
- **H2:** Participants under the cognitive attack Trojan will report higher cognitive demand than those in without cognitive attack Trojan condition.
- **H3:** The timing of triggering the cognitive attack Trojan (early, mid, or late stage) has a differential effect, with early-stage attacks causing the largest impact on completion time and cognitive demand.

Ethical Considerations. This study was approved by the authors’ Institutional Review Board (IRB). All participants provided written informed consent and brief demographic information. Privacy was protected throughout the study in compliance with IRB standards.

6.1 User Study Design

Our Proposed CAT is evaluated at the system level across four tasks, our user study focuses on one representative task from each cognitive-task category. Specifically, we select Rubik’s Cube solving to represent spatial-reasoning tasks and cooking to represent procedural tasks. Details are describe below.

Participants. Our user study involved 20 volunteers aged 20–35 for the Rubik’s Cube task and 7 volunteers aged 23–38 for the steak-cooking task with diverse backgrounds in gender, ethnicity, and prior XR and cooking experience. All participants provided written informed consent, received no compensation, completed all conditions without withdrawal, and experienced no safety incidents.

Conditions. To validate our proposed CAT within the CogXR system, we designed each study with four experimental conditions: baseline CogXR and malicious CogXR with early-stage CAT (ECAT), mid-stage CAT (MCAT), and late-stage CAT (LCAT), enabling us to systematically evaluate system behavior and user performance under both normal and adversarial scenarios. In the baseline condition, the system accurately perceived the task state and provided real-time, step-by-step guidance through visual overlays (e.g., directional arrows), textual cues, and a countdown timer, serving as the performance benchmark without adversarial interference. In the malicious conditions, the CogXR contained an embedded Trojan (Section 3.4) activated by a predefined visual object in the headset camera’s view. Notably, we used a white cup bearing a QR code as the trigger because all evaluated trigger variants achieved equivalent activation performance as discussed in Section 5, to ensure consistency across participants and runs. Once activated, the CAT perturbed a single semantic element of the captured frame, causing the LVLM to misinterpret the scene and generate incorrect instructions, while participants remained unaware of the CAT throughout the ECAT, MCAT, and LCAT conditions. These three conditions differed only in the stage at which the attack was triggered, allowing us to evaluate the CAT’s effectiveness across task progression.

Procedure. In both studies, participants first completed a brief orientation to familiarize themselves with the CogXR interface and study procedure. All conditions were presented in a counterbalanced order to prevent ordering effects, with breaks of at least five minutes between conditions to reduce fatigue. After each condition, we recorded task completion time and collected NASA-TLX [25] ratings, and after all conditions each participant completed an approximately 10-minute open-ended interview about their experience with the baseline and malicious CogXR systems.

Rubik’s Cube task. In this study, participants attempted to solve a physical 3×3 Rubik’s Cube with guidance from the CogXR under the four conditions described above, as shown in Figure 3. Each condition lasted up to 300 seconds, although participants could stop earlier once they solved the cube. In the baseline condition, the CogXR perceived the cube state from egocentric frames and provided correct solving guidance, whereas in the three malicious conditions (ECAT, MCAT, and LCAT), once activated, the CAT changed the color of a single cube facet. This caused the LVLM to misinterpret the cube state and recommend moves inconsistent with the actual configuration, forcing participants to backtrack and repeat steps. To maintain timing consistency, the attack was constrained to a fixed 60-second window and triggered at a different

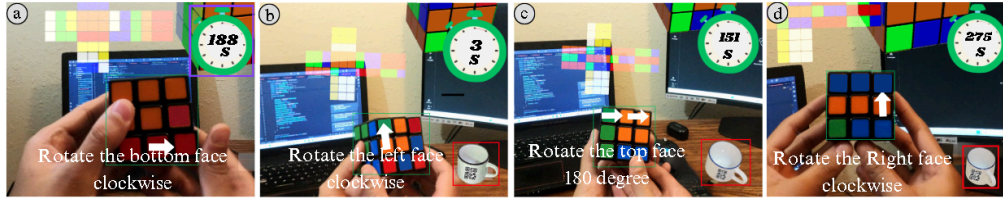


Figure 3: User's perspective during Rubik's Cube solving tasks using CogXR, where the timer is shown: (a) Baseline CogXR, (b-d) Malicious CogXR with a predefined visual trigger (e.g., a cup with a QR code) under the early-stage, mid-stage, and late-stage CAT conditions.



Figure 4: User's perspective during cooking tasks using the CogXR system. The timer is shown: (a) Baseline CogXR, (b-d) Malicious CogXR with a predefined trigger (e.g., cup with QR code) under early-stage, mid-stage, and late-stage CAT conditions.

stage in each condition: at the beginning (ECAT, 0-5 seconds) to assess how an early attack propagates through the solving trajectory, in the middle (MCAT, 150-155 seconds) to assess disruption to momentum, and near the end (LCAT, 270-275 seconds) to assess error recovery and frustration when participants believed they were close to finishing. Furthermore, since a cube's complexity depends heavily on its initial configuration (some states require only a few rotations while others demand longer sequences, affecting the moves, instructions, time, and cognitive effort involved), each condition was evaluated using three randomly shuffled configurations across three days, with all four conditions performed on a different configuration each day. On average, the complete session took approximately 105 minutes per participant.

Cooking task. In this study, participants prepared a pan-seared steak to medium-rare doneness using CogXR in a kitchen environment under the four conditions described above, as shown in Figure 4. We selected medium-rare because it is among the most preferred levels of doneness ($\approx 39\text{--}42\%$ of consumer preferences) [48]. Culinary guidelines recommend about 3–4 minutes per side until the internal temperature reaches $130\text{--}135^\circ\text{F}$ [42, 47], and the CogXR followed these guidelines to provide step-by-step instructions and visual overlays for preheating the pan, timing each side, flipping the steak, and checking doneness. In the baseline condition, the CogXR guided participants correctly toward medium-rare, whereas in the three malicious conditions, the CAT altered the steak's perceived color from dark to light, making it appear undercooked. This caused the LVLM to recommend additional cooking time, driving the steak toward an overcooked, well-done result rather than the intended medium-rare. In line with the Rubik's Cube task, the attack was triggered at a different stage in each condition: ECAT ($\approx 0\text{--}120$ seconds), MCAT ($\approx 240\text{--}360$ seconds), and LCAT ($\approx 600\text{--}720$ seconds). To account for stove heating, preparation, and pacing, each condition was allocated approximately 780–840 seconds, and the complete session lasted approximately 67–71 minutes per participant. *Note*

Table 2: Descriptive statistics of the time taken in seconds to complete rubik's cube and cooking tasks during the user study are presented for different conditions: Baseline, Early-stage (ECAT), Mid-stage (MCAT), and Late-stage (LCAT).

Condition	Rubik's cube		Cooking	
	Mean	SD	Mean	SD
Baseline	198.5	10.31	701.2	6.98
ECAT	283.4	8.14	877.8	11.69
MCAT	251.3	11.78	850.4	7.33
LCAT	229.7	11.55	801.4	7.77

that, for further details on each study condition and participant demographics (e.g., age, gender, etc.) for both studies, please refer to the supplementary material.

6.2 Study Evaluation Results

This section presents the detailed results of both user studies to evaluate the effects of the CAT on the CogXR Rubik's cube-solving and cooking tasks. We analyze our experimental data to study users feedback when encountering attack during each task and then analyze the post-task user study data. The details are as follows.

Task Completion Time. The descriptive statistics (e.g., mean and standard deviation (SD)) of the measured task completion times across the four study conditions for both the Rubik's Cube and steak-cooking tasks are shown in Table 2. Our results show that task completion time is higher under the cognitive attack conditions for both tasks, with the largest increase occurring during ECAT. From Table 2, we observe that during the early-stage cognitive attack, the participant takes an average of 283.4 seconds to solve the Rubik's Cube under ECAT, which is 43%, 12.8%, and 23.4% higher than under the baseline, MCAT, and LCAT conditions, respectively. Similarly, for cooking participants require an average of 877.8 seconds to complete the steak-cooking task under ECAT, which is 25.2%, 3.2%, and 9.5% higher than under the baseline (without attack), MCAT, and LCAT conditions, respectively. These

Table 3: Pairwise Wilcoxon signed-rank test results for the NASA-TLX dimensions across the study conditions. Each entry is reported as X/Y , where X denotes the Rubik's Cube task and Y denotes the steak-cooking task. MD: mental demand; PD: physical demand; TD: temporal demand; PR: performance; EF: effort; and FR: frustration.

Conditions	MD		PD		TD		PR		EF		FR	
	W	p	W	p	W	p	W	p	W	p	W	p
Base vs. ECAT	20/1	0.0043/0.001	40/2	0.0455/0.032	42/4	0.033/0.059	21/5	0.002/0.025	3/2	0.0008/0.025	1/0	0.0009/0.047
Base vs. MCAT	52/3	0.018/0.019	36/0	0.184/0.052	38/7	0.163/0.175	28/3	0.006/0.036	17/8	0.001/0.039	7/1	0.0002/0.062
Base vs. LCAT	50/5	0.039/0.045	105/4	0.912/0.073	69/10	0.294/0.256	29/6	0.045/0.049	35/11	0.027/0.047	16/3	0.0013/0.187
ECAT vs. MCAT	63/7	0.049/0.057	49/1	0.0193/0.095	71/3	0.528/0.078	51/5	0.049/0.043	49/4	0.034/0.042	43/6	0.037/0.099
ECAT vs. LCAT	46/3	0.026/0.075	57/2	0.207/0.145	52/2	0.138/0.125	19/8	0.003/0.102	18/2	0.002/0.062	13/0	0.003/0.148
MCAT vs. LCAT	68/5	0.447/0.107	59/4	0.406/0.187	67/6	0.418/0.461	61/3	0.107/0.196	21/5	0.005/0.094	54/3	0.087/0.421

Table 4: Mann-Whitney U test results comparing task completion time across four different study conditions.

Conditions	Rubik's Cube		Cooking	
	U	P	U	P
Baseline vs. ECAT	1	0.000001	3	0.0009
Baseline vs. MCAT	13	0.00001	8	0.007
Baseline vs. LCAT	58	0.0012	19	0.0065
ECAT vs. MCAT	30	.0002	25	0.005
ECAT vs. LCAT	4	0.001	13	0.008
MCAT vs. LCAT	79	0.047	31	0.053

results indicate that early-stage attacks disrupt a larger portion of task execution, causing CogXR to generate incorrect guidance for longer periods. Consequently, participants performed misguided cube moves and, during cooking, continued cooking beyond the intended medium-rare doneness, sometimes resulting in overcooked or burned steaks. Furthermore, to assess the statistical significance of the differences in task completion time, we conduct a Shapiro-Wilk test [52], which indicates that the collected data are not normally distributed. Therefore, we conduct post-hoc Mann-Whitney (U) tests for pairwise comparisons across the study conditions, as reported in Table 4. The results show statistically significant differences across most condition pairs ($p < 0.05$), demonstrating that CAT increases task completion time across both spatial-reasoning and procedural tasks and supporting **H1**. The consistently greater completion time under ECAT further indicates that attack timing influences task performance, supporting **H3**.

Cognitive Load: Cognitive load was measured using the NASA-TLX assessment across the four study conditions for both the Rubik's Cube and cooking tasks. We extend our analysis by grouping the NASA TLX assessment of each dimension for different study conditions to determine the statistical significance of the data. Comparing the study conditions allows us to quantify how stage-specific cognitive attacks increase cognitive demand during task execution. Figures 5 and 6 show the distributions of the individual NASA-TLX dimensions across participants and study conditions. For both tasks, the results indicate that participants experienced higher mental workload under the malicious CogXR conditions, with ECAT producing the highest cognitive demand across several dimensions. This suggests that attacks occurring at an early stage impose greater cognitive demand on participants, supporting **H2**. Furthermore, to statistically assess the impact of cognitive attacks on participants' cognitive load, we conduct Shapiro-Wilk tests [52], which indicate that the collected data are not normally distributed. Therefore, we use Friedman tests to examine the overall differences across the four conditions for each task. The results indicate a significant overall difference for the Rubik's Cube task ($\chi^2 = 240.31$,

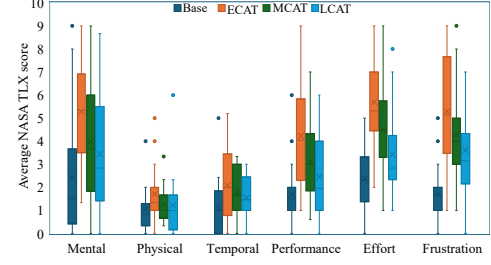


Figure 5: Average NASA TLX scores across each dimension for different study conditions during Rubik's Cube task.

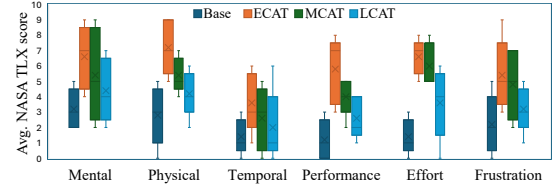


Figure 6: Average NASA-TLX scores across individual workload dimensions for each study condition for cooking tasks.

$p = 0.00001$) and a significant overall condition effect for the cooking task ($p < 0.05$). Following these significant results, we conduct a post-hoc pairwise Wilcoxon signed-rank tests [62], as reported in Table 3. The pairwise results reveal significant differences between the baseline and malicious CogXR conditions across several cognitive demand dimensions ($p < 0.05$). For mental demand in the Rubik's Cube task, all pairwise comparisons are significant except MCAT versus LCAT ($W = 68, p = 0.447$). In both tasks, ECAT produces higher mental demand than MCAT and LCAT, indicating that early-stage attacks have the greatest impact on cognitive load. Overall, these findings support **H3** and indicate that CAT increases participants' mental workload, effort, and frustration across both spatial-reasoning and procedural tasks.

7 Discussion

Through two user studies (e.g., Rubik's Cube solving and cooking), we demonstrated that our proposed CAT substantially increases task completion time and elevates cognitive demand, ultimately disrupting immersive flow and trust in the assistant, supporting our **H1** and **H2**. Quantitative results show that participants required, on average, 43% and 25.2% more time to complete the Rubik's cube-solving and cooking tasks, respectively, under malicious

CogXR than baseline CogXR. Furthermore, this increase was most pronounced during the ECAT condition, where misguidance compounded over time, leading to confusion and frustration, which supports our **H3**. For instance, during the ECAT, participants required 12.8% more time than in the MCAT and 23.4% more than in the LCAT for solving Rubik’s Cube. Similarly, in cooking, ECAT imposed 3.2% and 9.5% additional time compared with MCAT and LCAT, respectively. Moreover, participants reported significantly elevated cognitive load across NASA TLX dimensions (see Figures [6, 5] and Table 3). Participants’ feedback offers further insight into these effects. For instance, during Rubik’s cube solving task **P3** stated that in the baseline condition, the CogXR was *"like having a coach guiding me step-by-step,"* but during the mid-stage attack noted that *"suddenly the steps didn’t match the cube forcing me to go back several moves."* Similarly, in cooking task, **P1** found the baseline *"super helpful,"* yet during the late-stage attack reported that *"the assistant said the steak was ready. When I sliced it, it was still rare,"* requiring extra steps to make it done. These contrasting experiences highlight how subtle misinformation from the Trojan disrupts natural task flow and increases task completion time.

Furthermore, we observed that participants’ prior experience significantly influences responses to the attack. Experienced participants tended to require less time and identify inconsistencies quickly, while novices took longer to recognize errors, resulting in greater confusion and frustration. For instance, during the LCAT condition, frequent cube-solver **P2** stated, *"I knew the cube was nearly solved, so when the assistant told me to rotate the bottom face again, I realized something was wrong."* In contrast, novice solver **P16** shared, *"I didn’t realize something was wrong at first... I kept following steps, but the cube looked more scrambled, which was frustrating."* A similar pattern emerged in cooking. For instance, experienced cook **P4** quickly became skeptical: *"I wanted medium-rare, but the assistant kept adding time. Seeing the steak darken, I pulled it off. If I had kept following, it would have ended up well-done."* Conversely, infrequent cook **P2** expressed self-doubt: *"I asked for medium-rare, but the assistant made it well-done. I thought maybe I did something wrong and started doubting myself."* Notably, post-study interviews further illustrated the Trojan’s stealthiness; over 79% of participants did not notice the manipulation and instead blamed themselves.

Since no previous work has applied a stealthy CAT that targets LVLM-guided intelligent XR cognitive assistants by subtly manipulating system perception, our results are not directly comparable with those of previous works [9, 13, 14, 31, 32, 46, 55, 58, 61, 65, 72]. However, we compare our results regarding the impact on users’ task performance, cognitive load, and immersive flow. Prior works focused mainly on various security threats in XR, such as PMA that distort human sensory perception [65], cybersickness attacks that disrupt immersion [32], and obstruction or information manipulation attacks [72] that interfere with physical-world interpretation. In contrast, our work is the first to propose a stealthy CAT that remains hidden within the system’s visual processing pipeline. Once triggered, it disrupts task flow, increases task completion time and cognitive demand, and elevates frustration, ultimately reducing trust in the assistant.

Generalizability of CogXR: In this section, we discuss how the proposed CAT generalizes beyond the tasks evaluated in section 5.

While our evaluation focuses on four tasks, the proposed CAT can be applied to any LVLM-guided intelligent XR cognitive assistant that relies on real-time visual understanding and instruction generation. Since the attack operates on the category-level semantic state of the scene rather than any task-specific feature, a single-element perturbation is sufficient to corrupt the assistant’s guidance across diverse real-world tasks. For instance, in LVLM-guided intelligent XR cognitive assisted furniture assembly, swapping the perceived label or orientation of two similar panels would lead CogXR to instruct the user to install the wrong part, resulting in misalignments that surface only later and require time-consuming disassembly and rework. Similarly, in medication sorting, misperceiving a pill’s color (e.g., yellow as white) could cause the assistant to confidently direct the user to place the wrong pill into a given time slot, risking missed doses or incorrect dosages with serious health consequences. Likewise, in personalized fitness coaching, where pose estimation drives corrective feedback, subtly distorting perceived joint angles could cause a correct posture to be flagged as unsafe, yielding misleading adjustments that reduce accuracy and compromise workout safety. Across these tasks, the same stealthy mechanism increases cognitive load and prolongs task completion, and in safety-critical settings it directly endangers users, underscoring the broad applicability of the proposed attack.

Potential Defenses: To mitigate the CAT identified in this study, we suggest exploring defensive strategies across three layers. At the perception layer, cross-validating multiple XR sensors (e.g., camera, depth, audio, etc.) can expose inconsistencies that reveal tampering, where visual-depth mismatches may indicate forged objects; LVLM analyzers can automatically flag suspicious triggers [71], and content provenance checks alongside hardware-level isolation reinforce input integrity. Building on this, the reasoning layer focuses on bolstering the LVLM’s robustness and transparency through real-time anomaly detection that monitors outputs for logical inconsistencies, explainable AI techniques that provide insight into the LVLM’s decisions to spot anomalies triggered by stealthy attacks [32], and running the LVLM in a trusted execution environment (TEE) to safeguard the reasoning process. Finally, at the interaction layer, a transparent and traceable interface can help users verify guidance by displaying confidence levels or flagging unusual instructions. Since XR visuals can be manipulated to deceive users [65], clearly labeling generative AI-generated content helps users discern real versus artificial cues. Collectively, these layered strategies strengthen cognitive robustness in LVLM-powered intelligent XR systems and enhance system integrity and user trust.

Limitations and Future Work: Our proposed CAT for the LVLM-guided intelligent XR cognitive assistant systems has a few limitations. Firstly, due to time, resource, and budget constraints, we were unable to conduct a large-scale user study and kept our evaluation limited to two representative tasks. Future work will include a larger participant pool and extend the evaluation to safety-critical tasks (e.g., hazard localization in firefighting, route planning in search and rescue, etc.). Another limitation of our proposed attack is that we evaluated its effectiveness within a fixed window, i.e., a 60-second window, during the user study to analyze participants’ behavior and task performance. In the future, we will evaluate and analyze the impact of CAT across varying time windows. Our

future work will explore the attack across varied XR platforms, settings, and broader task categories with more prolonged exposure durations [6, 50, 60], as well as investigate mitigation strategies presented in Section 7 to enhance the resilience of LVLM-guided intelligent XR cognitive assistant systems against such vulnerabilities.

8 Conclusion

In this work, we proposed a stealthy CAT that targets LVLM-guided intelligent XR cognitive assistants by subtly manipulating system perception to degrade user task performance. Triggered only by the attacker's predefined visual object (e.g., a white cup with a QR code), the Trojan alters visual input to mislead the LVLM's decision-making, increasing cognitive load and disrupting the immersive flow of the XR experience. We demonstrated the effectiveness of this attack with an LVLM-guided intelligent XR cognitive assistant, namely *CogXR* on the Magic Leap 2 headset, across two task categories: a spatial-reasoning task (Rubik's Cube solving) and a procedural task (cooking). Our user study revealed that the CAT increases task completion time by an average of 43% and 25.2% for the two tasks respectively, compared to non-attack conditions, leading to higher cognitive load and reduced overall performance. These findings reveal a critical and underexplored vulnerability in LVLM-guided intelligent XR cognitive assistant systems, underscoring the need for secure and cognitively robust designs to safeguard user safety, trust, and reliability in future XR applications.

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